Math-4141

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Pair of Straight Lines

Overview of a Pair of Straight Lines

When the product of two linear equations in x and y indicates a straight line is multiplied together, a pair of straight lines is generated.

Let , $L_1=0$ and $L_2=0$ be two straight line equations.

If $P(x_1,y_1)$ is a point on L_1, the equation $L_1=0$ is satisfied. If $P(x_1,y_1)$ is a point on $L_2=0$, then the equation is satisfied.

$P(x_1,y_1)$ satisfies the equation $L_1L_2=0$ if it is located on L_1 or L_2.

∴

$L_1L_2=0$ denotes the pair of straight lines $L_1=0$ and $L_2=0$, and $L_1L_2=0$ denotes the joint equation of $L_1=0$ and $L_2=0$.

We get an equation of the type

$ax^2+2hxy+by^2+2gx+2fy+c=0$ when we extend the above equation. This is a non-homogeneous second degree equation in x and y.

If a,b,h are not all zero, the general equation of a second degree homogeneous equation in x and y is $ax^2+2hxy+by^2=0$.

$ax^2+2hxy+by^2=0$,a pair of straight lines that pass through the origin

If a,b,h are not all zero, the general equation of a second-degree non-homogeneous equation in x and y is $ax^2+2hxy+by^2+2gx+2fy+c=0$.

Formulas for a Pair of Straight Lines

The following is a list of pair of straight lines formulas:

1. $ax^2+2hxy+by^2=0$ is a second-degree homogeneous equation that depicts a pair of straight lines flowing through the origin.

- As a result, if $h^2>ab$, the two straight lines are distinct and real
- The two straight lines are coincident if $h^2=ab$
- If $h^2<ab$, the two straight lines with the origin as the point of intersection are imaginary

* By choosing a, b, h verify the formulas and hence draw using Desmos.

Example 6.34 If exists, find the straight lines by separating the equations $2x^2 + 2xy + y^2 = 0$.

Solution:
Since the given equation is a homogeneous equation, divide the given equation $2x^2 + 2xy + y^2 = 0$ by x^2 and substituting $\frac{y}{x} = m$

We get $m^2 + 2m + 2 = 0$

The values of m (slopes) are not real (complex number), therefore no line will exist with the joint equation $2x^2 + 2xy + y^2 = 0$

We sometimes say that this equation represents imaginary lines.

Note that in the entire plane, only $(0, 0)$ satisfies this equation

Angle Formed by Two Straight Lines

Consider the equation

$ax^2+2hxy+by^2=0$ .....(1)

for a pair of straight lines passing through the origin.

Let the slopes of these two lines be m_1 and m_2. By y dividing (1) by x^2 and substituting $y/x=m$, we get

$bm^2+2hm+a=0$

The roots of this quadratic in m will be m_1 and m_2. As a result, $m_1+m_2= -2h/b$ and $m_1m_2=ab$.

The angle formed by the two lines is θ

Then,

$\tan\theta=|m_2-m_1|/|1+m_2m_1|$

$=|\sqrt{(m_1+m_2)^2-4m_1m_2}/\sqrt{1+m_2m_1}|$

$=|\sqrt{(-2h/b)^2-4a/b}/|\sqrt{1+ab}|$

$\tan\theta=|2\sqrt{h^2-ab}/a+b|$

Pair of Straight Lines Condition

When dealing with difficulties based on a pair of straight lines, some key conditions or outcomes are particularly useful.

The following are the outcomes of a general second-degree equation $ax^2+2hxy+by^2+2gx+2fy+c=0$, which represents a pair of straight lines.

1.If $\tan\theta=0$, two lines will be parallel or coincident. i.e. if $h^2-ab=0$

2. If $\tan\theta$ is not defined, $a+b=0$, two lines will be perpendicular.

3. If the coefficient of $xy=0$, i.e. if $h=0$, two lines will be equally inclined to the axes.

4. The angle formed by two straight lines is given by $\tan\theta=|2\sqrt{h^2-ab}|/|a+b|$.

5. If $ax^2+2hxy+by^2+2gx+2fy+c=0$ denotes a pair of straight lines, the sum of their slopes is $-2h/b$ and their product is a/b.

Example 6.36 Show that the straight lines $x^2 - 4xy + y^2 = 0$ and $x + y = 3$ form an equilateral triangle.

Solution:

Let the line $x + y = 3$ intersects the pair of line x^2 -$4xy + y^2 = 0$ at A and B.

The angle between the lines $x^2 - 4xy + y^2 = 0$ is

$$\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a+b}\right| = \frac{2\sqrt{4-1}}{2} = \sqrt{3}$$
$$\Rightarrow \theta = \tan^{-1}\sqrt{3} = 60^\circ$$

The angle bisectors of the angle AOB are given by

$$\frac{x^2 - y^2}{a-b} = \frac{xy}{h}$$
$$\Rightarrow x^2 - y^2 = 0$$
$$\Rightarrow x + y = 0 \quad \text{and} \quad x - y = 0$$

The angle bisectors $x - y = 0$ is perpendicular to the given line through AB,

that is $x + y = 3 \Rightarrow \Delta OAB$ is isosceles.
$$\Rightarrow \angle ABO = \angle BAO = 60^\circ$$

therefore the given lines form an equilateral triangle.

3. Equation of the bisectors of the angle between the lines

$ax^2 + 2hxy + by^2 = 0$ Let the equations of the two straight lines be $y - m_1x = 0$ and $y - m_2x = 0$

$$\therefore\ m_1 + m_2 = -\frac{2h}{b} \quad \text{and} \quad m_1m_2 = \frac{a}{b}$$

We know that the equation of bisectors is the locus of points from which the perpendicular drawn to the two straight lines are equal.

Let $P\ (p, q)$ be any point on the locus of bisectors.

The perpendiculars from $P\ (p, q)$ to the line $y - m_1x = 0$ is equal to the perpendicular from $P\ (p, q)$ to $y - m_2x = 0$

$$\pm\frac{q - m_1p}{\sqrt{1 + m_1^2}} = \pm\frac{q - m_2p}{\sqrt{1 + m_2^2}}$$

That is, $$(q - m_1p)^2\,(1 + m_2^2) = (q - m_2p)^2\,(1 + m_1^2)$$

Simplifying we get $$p^2 - q^2 = 2pq\left(\frac{1 - m_1m_2}{m_1 + m_2}\right)$$

$$\Rightarrow p^2 - q^2 = 2pq\left(\frac{1 - \frac{a}{b}}{-\frac{2h}{b}}\right)$$

That is $$\frac{p^2 - q^2}{a - b} = \frac{pq}{h}$$

$\therefore$ The locus of $P(p, q)$ is $$\frac{x^2 - y^2}{a - b} = \frac{xy}{h} \qquad (6.36)$$

Example 6.37 If the pair of lines represented by $x^2 - 2cxy - y^2 = 0$ and $x^2 - 2dxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, prove that $cd = -1$.

Solution:

Given that the pair of straight lines,

$$x^2 - 2cxy - y^2 = 0 \tag{6.37}$$
$$x^2 - 2dxy - y^2 = 0 \tag{6.38}$$

The equation of the angle bisectors of (6.37) is

$$\frac{x^2 - y^2}{2} = \frac{xy}{-c}$$
$$\Rightarrow \quad cx^2 + 2xy - cy^2 = 0 \tag{6.39}$$

Given that the angle bisector of (6.37) is (6.38)
Therefore equations

$$x^2 - 2dxy - y^2 = 0$$
$$cx^2 + 2xy - cy^2 = 0$$

represent the same equation as the angle bisector of (6.37)

Comparing the like terms of the above two equations, we get

$$\frac{1}{c} = \frac{-2d}{2} = \frac{-1}{-c}$$
$$\Rightarrow \quad cd = -1$$

4. General form of Pair of Straight Lines

Consider the equations of two arbitrary lines $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$ The combined equation of the two lines is

$(l_1x + m_1y + n_1)\,(l_2x + m_2y + n_2) = 0$

If we multiply the above two factors together, we get a more general equation to a pair of straight lines has the form

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ 6.40

The above equation is a non homogenous equation of degree two.

in which

$$a = l_1l_2 \qquad 2g = l_1n_2 + l_2n_1$$
$$b = m_1m_2 \qquad 2f = m_1n_2 + m_2n_1$$
$$c = n_1n_2 \qquad 2h = l_1m_2 + l_2m_1$$

An equation of the form (6.40) will always represent a pair of straight lines, provided it must able to be factorized into two linear factors of the form $l_1x + m_1y + n_1 = 0$ and $l_2x + m_2y + n_2 = 0$.

Condition that the general second degree equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ should represent a pair of straight lines

Let us rearrange the equation of the pair of straight lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ as a quadratic in x, we have

$$ax^2 + 2(hy + g)x + (by^2 + 2fy + c) = 0$$

$$x = \frac{-(hy + g) \pm \sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}}{a}$$

$$ax + hy + g = \pm\sqrt{(hy + g)^2 - a(by^2 + 2fy + c)}$$

That is, $ax + hy + g = \pm\sqrt{(h^2 - ab)y^2 + 2(gh - af)y + g^2 - ac}$

Since each of the above equations represents a straight line, they must be of the first degree in x and y. Therefore the expression under the radical sign should be a perfect square and the condition for this is

$$4(gh - af)^2 - 4(h^2 - ab)(g^2 - ac) = 0$$

Simplifying and dividing by a,

We get, $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ or

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$ (Expansion of Determinant will be studied in the next chapter)

The relationship between the equations of pair of straight lines

$$ax^2 + 2hxy + by^2 = 0 \quad (6.41)$$

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0) \quad (6.42)$$

The slope of the above pair of straight lines (6.41) and (6.42) are depending only on the coefficients of

$$x^2,\ xy \text{ and } y^2.$$

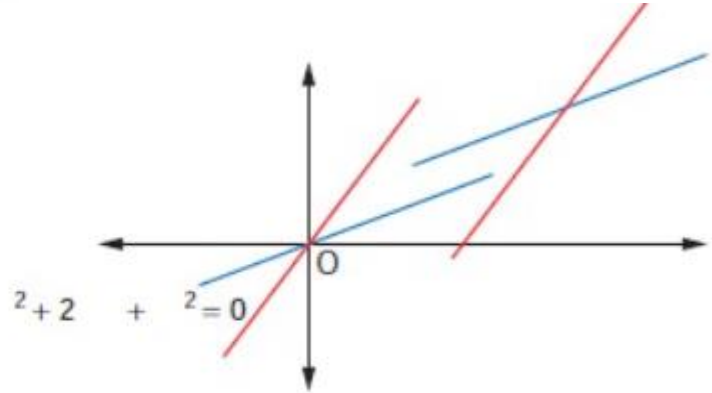


Figure 6.49

(i) If the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, then $ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines through the origin parallel to the first pair. The point of intersection (6.41) is (0, 0) and the point of intersection (6.42) is

$$P\left(\frac{hf - bg}{ab - h^2}, \quad \frac{gh - af}{ab - h^2}\right)$$

(ii) If θ be the angle between the straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then it will have the same value as the angle between the two lines represented by $ax^2 + 2hxy + by^2 = 0$

$$\text{Thus } \theta = \tan^{-1}\left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$$

(iii) If the two straight lines represented by the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ are at right angles, then the two lines represented by $ax^2 + 2hxy + by^2 = 0$ are also at right angles and the condition is $a + b = 0$

Example 6.38 If the equation $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, find (i) the value of λ and the separate equations of the lines (ii) point of intersection of the lines (iii) angle between the lines

Solution:

(i) general equation is $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

given equation is $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$

Comparing the given equation with the general equation of the second degree we have

$$a = \lambda, b = 12, c = -3, h = -5, g = \frac{5}{2}, f = -8$$

Now applying the condition for pair of straight lines

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\text{i.e. } \lambda(12)(-3) + 2(-8)\left(\frac{5}{2}\right)(-5) - \lambda(-8)^2 - 12\left(\frac{5}{2}\right)^2 - (-3)(-5)^2 = 0$$

$$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0, \quad \Rightarrow \lambda = 2$$

$\therefore$ The equation is $2x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$.
Let us first factorize the terms of second degree terms from the above equation,

$$\text{we get} \qquad 2x^2 - 10xy + 12y^2 \equiv (x - 2y)(2x - 6y)$$

$$\therefore \; 2x^2 - 10xy + 12y^2 + 5x - 16y - 3 \equiv (x - 2y + c_1)(2x - 6y + c_2)$$

Equating like terms, we get

$$2c_1 + c_2 = 5, \quad 3c_1 + c_2 = 8, \quad c_1c_2 = -3$$

Solving first two equation, we get $c_1 = 3, c_2 = -1$.

$\therefore$ The separate equations of the lines are

$$x - 2y + 3 = 0 \quad \text{and} \quad 2x - 6y - 1 = 0$$

(ii) Point of intersection of the lines is given by solving the two equation of the lines, we get

$$(x, y) = \left(-10, -\frac{7}{2}\right) \left(\text{or use the formula } \left(\frac{hf - bg}{ab - h^2}, \frac{hg - af}{ab - h^2}\right)\right)$$

(iii) Angle between the lines is given by

$$\begin{aligned} \tan\theta &= \left| \frac{2\sqrt{h^2 - ab}}{a + b} \right| \\ &= \frac{2\sqrt{25 - 24}}{2 + 12} = \frac{1}{7} \\ \therefore\ \theta &= \tan^{-1}\left(\frac{1}{7}\right) \end{aligned}$$

Example 6.39 A student when walks from his house, at an average speed of 6 kmph, reaches his school ten minutes before the school starts. When his average speed is 4 kmph, he reaches his school five minutes late. If he starts to school every day at 8.00 A.M, then find (i) the distance between his house and the school (ii) the minimum average speed to reach the school on time and time taken to reach the school (iii) the time the school gate closes (iv) the pair of straight lines of his path of walk.

Solution:

Let x-axis be the time in hours and y-axis be the distance in kilometer.
From the given information, we have

$$y = 6(x - \frac{10}{60}) \Rightarrow y = 6x - 1 \tag{6.43}$$

$$y = 4(x + \frac{5}{60}) \Rightarrow y = 4x + \frac{1}{3} \tag{6.44}$$

Solving the above two equations, we get $(x, y) = \left(\frac{2}{3}, 3\right)$

$$x = \frac{2}{3} \text{ hour} = 40 \text{ minutes}, \ y = 3 \text{ km}$$

(i) the distance between house and the school is 3km
(ii) the minimum average speed to reach the school on time is

$$\frac{60}{40} \times 3 = 4.5 \text{ kmph}$$

and time taken is hours $\frac{2}{3}$ or 40 minutes

(iii) the school gate closes at 8.40 AM
(iv) the pair of straight lines of his path of walk to school is

$$(6x - y - 1)\left(4x - y + \frac{1}{3}\right) = 0$$

$$72x^2 - 30xy + 3y^2 - 6x + 2y - 1 = 0 \tag{6.45}$$

Draw the graph by using the site: https://www.geogebra.org/graphing
https://www.geogebra.org/m/fhS6HUtP

Example 6.40 If one of the straight lines of $ax^2 + 2hxy + by^2 = 0$ is perpendicular to $px + qy = 0$, then show that $ap^2 + 2hpq + bq^2 = 0$.

Solution:

Let m_1 and m_2 be the slopes of the pair of lines $ax^2 + 2hxy + by^2 = 0$ and m be the slope of $px + qy = 0$

Therefore, $m_1 + m_2 = -\frac{2h}{b}$, $m_1 m_2 = \frac{a}{b}$ and $m = -\frac{p}{q}$

since one of the straight lines of $ax^2 + 2hxy + by^2 = 0$ is perpendicular to $px + qy = 0$,

$$\begin{aligned}
m\,m_1 = -1 \quad \text{or} \quad m\,m_2 &= -1 \\
\Rightarrow \quad (m\,m_1 + 1) = 0 \text{ or } (m\,m_2 + 1) &= 0 \\
\Rightarrow \quad (m\,m_1 + 1)(m\,m_2 + 1) &= 0 \\
\Rightarrow \quad (m_1 m_2)m^2 + (m_1 + m_2)m + 1 &= 0 \\
\Rightarrow \quad \left(\frac{a}{b}\right)\left(-\frac{p}{q}\right)^2 + \left(-\frac{2h}{b}\right)\left(-\frac{p}{q}\right) + 1 &= 0 \\
\Rightarrow \quad ap^2 + 2hpq + bq^2 &= 0
\end{aligned}$$

Example 6.41 Show that the straight lines joining the origin to the points of intersection of $3x - 2y + 2 = 0$ and $3x^2 + 5xy - 2y^2 + 4x + 5y = 0$ are at right angles

Solution:

The straight lines joining the origin and the points of intersection of given equations is a second degree homogeneous equation.

Following steps show, the way of homogenizing the $3x^2 + 5xy - 2y^2 + 4x + 5y = 0$ with $3x - 2y + 2 = 0$

$$3x^2 + 5xy - 2y^2 + (4x + 5y)(1) = 0 \text{ and } \frac{(3x - 2y)}{-2} = 1$$

$$3x^2 + 5xy - 2y^2 + (4x + 5y)\left(\frac{3x - 2y}{-2}\right) = 0$$

$$(-2)(3x^2 + 5xy - 2y^2) + (4x + 5y)(3x - 2y) = 0$$

On simplification,

We get, $2x^2 + xy - 2y^2 = 0. \quad \Rightarrow a = 2, b = -2 \Rightarrow a + b = 0$

Since sum of the co-efficient of x^2 and y^2 is equal to zero, the lines are at right angles

- **Solve all examples from the book:**
  Co-ordinate Geometry with Vector Analysis - Rahman & Bhattacharjee.